Assignment – 11

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SECTION A - CONSTRUCTORS

Q1) Create a Java class named Student with the following requirements: Declare the following instance variables:

- **Student ID**
- **Student Name**
- **Course**
- **Age**

Implement:

- **A default constructor that initializes default values.**
- **A parameterized constructor that accepts all four values.**
- **A method displayDetails() to display student information.**

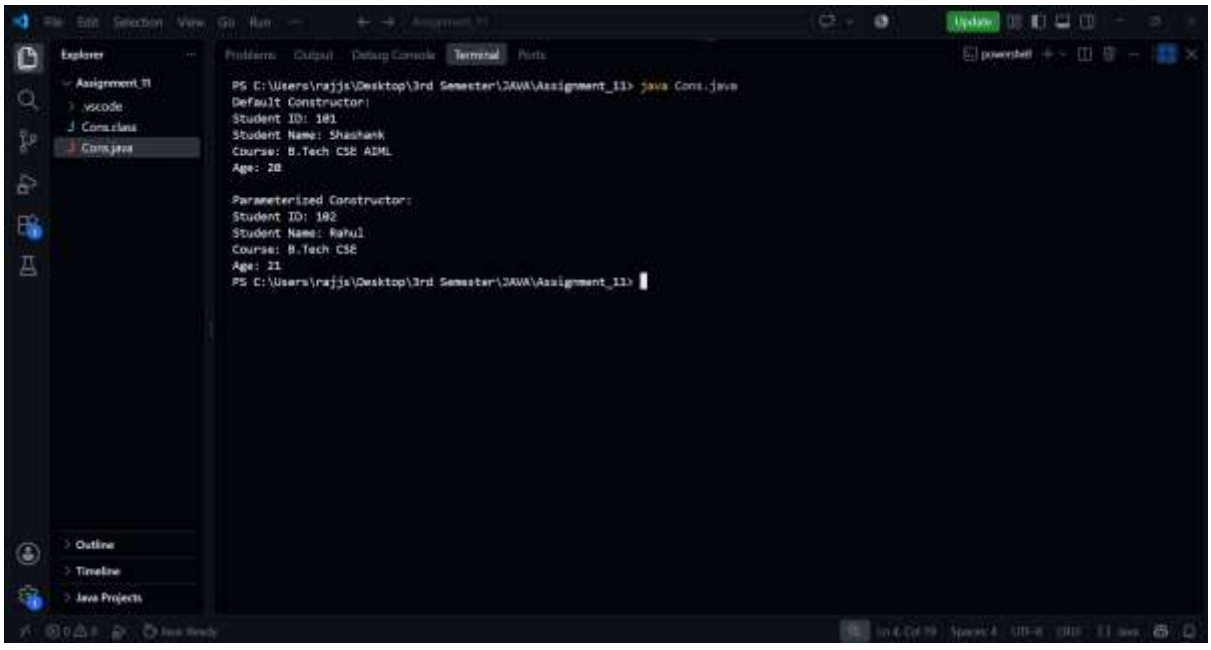
Create a Main class and demonstrate both constructors.

Answer-

```
class Student {
   int studentId;
   String studentName;
   String course;
   int age;
   Student() {
      studentId = 101;
      studentName = "Shashank";
      course = "B.Tech CSE AIML";
      age = 20;
   }
   Student(int id, String name, String course, int age) {
      this.studentId = id;
      this.studentName = name;
      this.course = course;
```

```
        this.age = age;
    }
    void displayDetails() {
        System.out.println("Student ID: " + studentId);
        System.out.println("Student Name: " + studentName);
        System.out.println("Course: " + course);
        System.out.println("Age: " + age);
    }
}
public class Cons{
    public static void main(String[] args) {
        Student s1 = new Student();
        System.out.println("Default Constructor:");
        s1.displayDetails();
        Student s2 = new Student(102, "Rahul",
                "B.Tech CSE", 21);
        System.out.println("\nParameterized Constructor:");
        s2.displayDetails();
    }
}
```

OUTPUT-



Q2) Create a class named Employee with the following requirements:

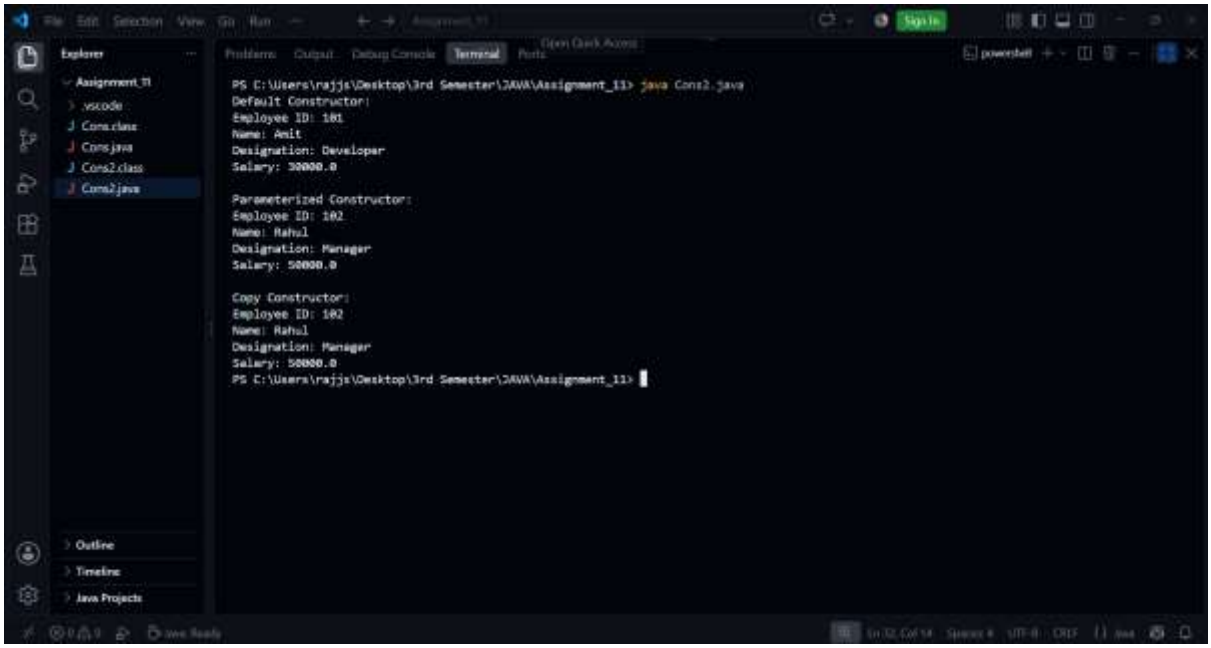
- **Declare employee ID, name, designation, and salary.**
- **Create a default constructor.**
- **Create a parameterized constructor.**
- **Create a copy constructor that initializes an object using another Employee object.**
- **Display the details of all three employee objects**

Answer-

```
class Employee {
   int employeeId;
   String name;
   String designation;
   double salary;
   Employee() {
      employeeId = 101;
      name = "Amit";
      designation = "Developer";
      salary = 30000;
   }
   Employee(int id, String name,
         String designation, double salary) {
      this.employeeId = id;
      this.name = name;
      this.designation = designation;
      this.salary = salary;
   }
   Employee(Employee e) {
      this.employeeId = e.employeeId;
      this.name = e.name;
      this.designation = e.designation;
      this.salary = e.salary;
   }
   void displayDetails() {
      System.out.println("Employee ID: " + employeeId);
```

```
        System.out.println("Name: " + name);
        System.out.println("Designation: " + designation);
        System.out.println("Salary: " + salary);
    }
}
public class Cons2 {
    public static void main(String[] args) {
        Employee e1 = new Employee();
        Employee e2 = new Employee(102, "Rahul",
                "Manager", 50000);
        Employee e3 = new Employee(e2);
        System.out.println("Default Constructor:");
        e1.displayDetails();
        System.out.println("\nParameterized Constructor:");
        e2.displayDetails();
        System.out.println("\nCopy Constructor:");
        e3.displayDetails();
    }
}
```

OUTPUT-



Section B – INHERITANCE

Q1) Create the following inheritance structure:

- **Parent Class: Person**
- **Name**
- **Age**
- **Email**
- **Method: displayPersonDetails()**

Child Class: Student

- **Student ID**
- **Course**
- **Method: displayStudentDetails()**

Requirements:

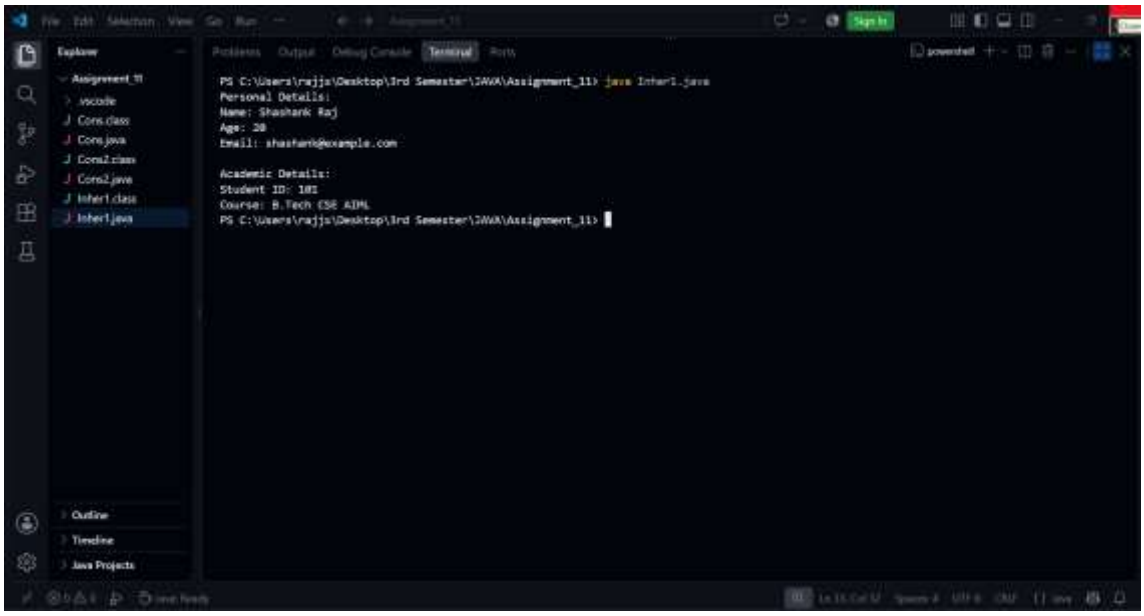
- **Implement single inheritance.**
- **Use the super keyword to call the parent class constructor.**
- **Create a Student object.**
- **Display both personal and academic details.**

Answer-

```
class Person {
   String name;
   int age;
   String email;
   Person(String name, int age, String email) {
      this.name = name;
      this.age = age;
      this.email = email;
   }
   void displayPersonDetails() {
      System.out.println("Name: " + name);
      System.out.println("Age: " + age);
      System.out.println("Email: " + email);
   }
}
class Student extends Person {
   int studentId;
   String course;
```

```
    Student(String name, int age, String email,
            int studentId, String course) {
        super(name, age, email);
        this.studentId = studentId;
        this.course = course;
    }
    void displayStudentDetails() {
        System.out.println("Student ID: " + studentId);
        System.out.println("Course: " + course);
    }
}
public class Inher1 {
    public static void main(String[] args) {
        Student s = new Student(
                "Shashank Raj",
                20,
                "shashank@example.com",
                101,
                "B.Tech CSE AIML"
        );
        System.out.println("Personal Details:");
        s.displayPersonDetails();
        System.out.println("\nAcademic Details:");
        s.displayStudentDetails();
    }
```

OUTPUT-



Q2) Create a Java program demonstrating multilevel inheritance.
Implement the following structure:
Employee
|
Manager
|
ProjectManager
Requirements:

- **Employee ID**
- **Employee Name**
- **Basic Salary**
- **Department**
- **Team Size**
- **Project Name**
- **Project Duration**

Create constructors in all three classes and use super() to invoke parent constructors.
Display complete project manager details.

Answer-

```
class Employee {
   int employeeId;
   String employeeName;
   double basicSalary;
   Employee(int employeeId, String employeeName,
         double basicSalary) {
      this.employeeId = employeeId;
      this.employeeName = employeeName;
      this.basicSalary = basicSalary;
   }
}
class Manager extends Employee {
   String department;
   int teamSize;
   Manager(int employeeId, String employeeName,
         double basicSalary, String department,
         int teamSize) {
```

```
        super(employeeId, employeeName, basicSalary);
        this.department = department;
        this.teamSize = teamSize;
    }
}
class ProjectManager extends Manager {
    String projectName;
    int projectDuration;
    ProjectManager(int employeeId, String employeeName,
                   double basicSalary, String department,
                   int teamSize, String projectName,
                   int projectDuration) {

        super(employeeId, employeeName, basicSalary,
              department, teamSize);

        this.projectName = projectName;
        this.projectDuration = projectDuration;
    }
    void displayDetails() {
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Employee Name: " + employeeName);
        System.out.println("Basic Salary: " + basicSalary);
        System.out.println("Department: " + department);
        System.out.println("Team Size: " + teamSize);
        System.out.println("Project Name: " + projectName);
        System.out.println("Project Duration: "
              + projectDuration + " months");
    }
}
public class Inher2 {
    public static void main(String[] args) {
        ProjectManager pm = new ProjectManager(
                101,
                "Shashank Raj",
                75000,
```

```
            "Information Technology",
            10,
            "AI Based Management System",
            12
        );
        System.out.println("Project Manager Details:");
        pm.displayDetails();
    }
}
```

OUTPUT-

